

Aerospace series**Bolt, Blind, 100° Countersunk Head**

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Contents

- 1 Scope
- 2 Normative references
- 3 Requirements
- 4 Designation
- 5 Marking
- 6 Technical specification

1 Scope

This standard specifies the dimensions, required characteristics and the mass of a high strength, 100° Countersunk head Blind Bolt for use in metallic applications.

2 Normative references

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS2320	Fastener, Blind, High Strength, General Technical Specification
AMS2700	Passivation of Corrosion Resistant Steels
AMS4928	Titanium Alloy Bars, Wire, Forgings, Rings, and Drawn Shapes 6Al - 4V Annealed
AMS4967	Titanium Alloy, Bars, Wire, Forgings, and Rings 6.0Al - 4.0V Annealed, Heat Treatable
AMS5732	Steel, Corrosion and Heat Resistant, Bars, Wire, Forgings, Tubing and Rings 15Cr 25.5Ni 1.2Mo 2.1Ti 0.006B 0.30V Consumable Electrode Melted 1800°F (982°C) Solution and Precipitation Heat Treated
AMS5737	Steel, Corrosion and Heat Resistant, Bars, Wire, Forgings and Tubing 15Cr 25.5Ni 1.2Mo 2.1Ti 0.006B 0.30V, Consumable Electrode Melted, 1650°F (899°C), Solution and Precipitation Heat Treated
ANSI/ASME-B46-1	Surface texture (surface roughness waviness, and lay).
AS5272	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting, Procurement Specification
AS87132	Lubricant, Cetyl Alcohol, 1-Hexadecanol, Application to Fasteners
ASNA2657	Bolt - reduced Countersunk head, shear coupling, Short thread
EN2424	Aerospace series - Marking of aerospace products
EN4473	Aerospace series - Aluminium pigmented coatings for fasteners - Technical specification
EN6114	Bolt - Countersunk head, Short thread Inch series
EN6117	Specification for lubrication of fasteners with cetyl alcohol
ISO2768-1	General tolerances; part 1: tolerances for linear and angular dimensions without individual tolerance indications
ISO2768-2	General tolerances; part 2: geometrical tolerances for features without individual tolerance indications

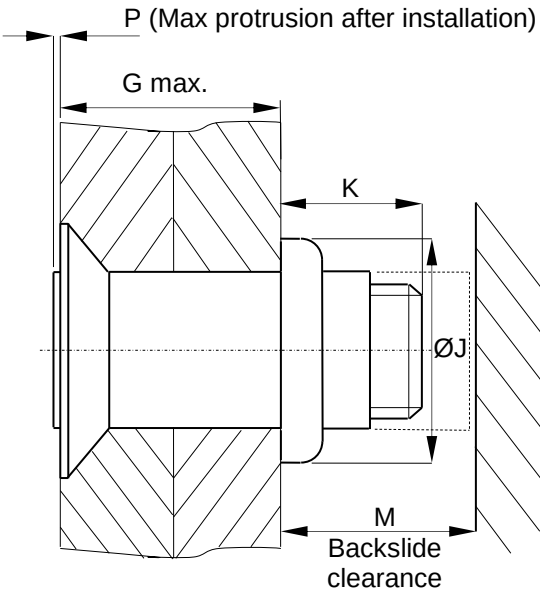
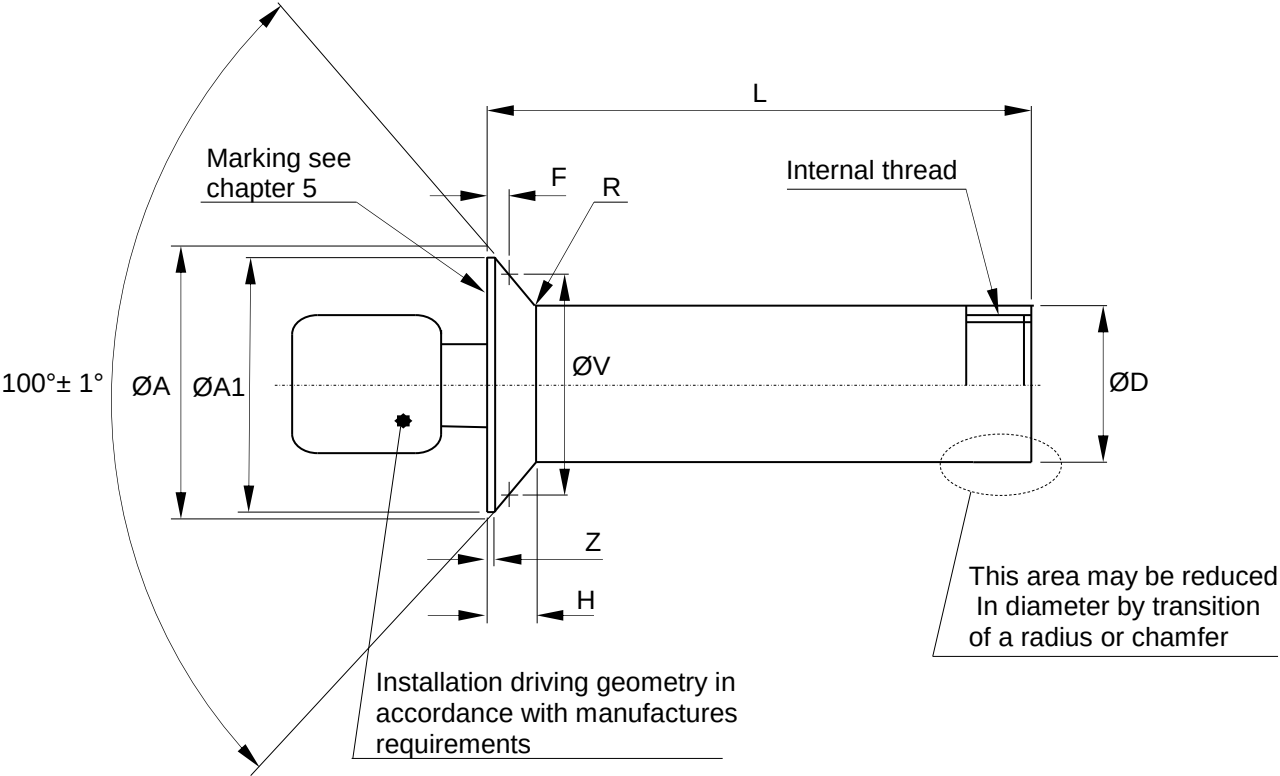
3 Requirements

3.1 Configuration, dimensions, tolerances and mass

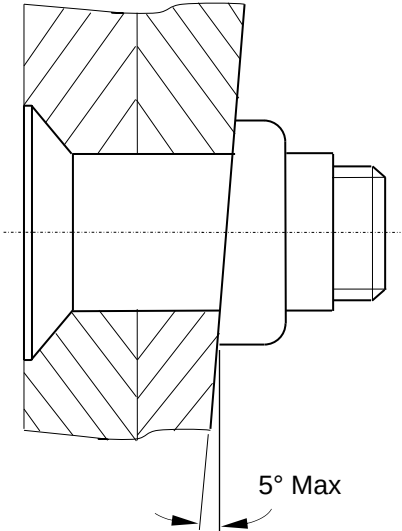
The configuration, dimensions and tolerances shall be in accordance with Figure 1 and Tables 1 and 2. First (1/64) and Second (1/32) oversize dimensions shall be in accordance with Tables 3 and 4. Material shall be in accordance with Table 5. Mechanical characteristics shall be in accordance with Table 6. Mass values shall be in accordance with Table 7. Tolerances not specified, shall be in accordance with ISO2768-1 and ISO2768-2. Dimensions apply after coating and before lubrication.

3.2 General characteristics

Surface condition as per ANSI-B46-1



Installation requirements



Maximum back side slope angle

Note: The ABS2322 shall include a thread locking feature.

Figure 1: Configuration

Table 1: Dimensions

Diameter code	ØA Theo Ref	ØA1 Min	ØD		F		H Ref	ØJ Min
			Max	Min	Max	Min		
5	.2877 (7,308)	.2598 (6,599)	.1635 (4,153)	.1625 (4,128)	.0330 (0,838)	.0298 (0,757)	.049 (1,24)	.205 (5,21)
6	.3277 (8,324)	.2929 (7,440)	.1895 (4,813)	.1885 (4,788)	.0295 (0,749)	.0263 (0,668)	.056 (1,42)	.238 (6,05)
8	.4283 (10,879)	0.393 (10,01)	.2495 (6,337)	.2485 (6,312)	.0227 (0,577)	.0195 (0,495)	.074 (1,88)	.313 (7,95)
10	.5361 (13,617)	0.501 (12,73)	.3120 (7,920)	.3110 (7,900)	.0234 (0,594)	.0198 (0,503)	.092 (2,34)	.391 (9,93)

Table 1: Dimensions (concluded)

Diameter code	K Max	M Ref	P Max	R Max	ØV		Z Max
					Max	Min	
5	0.409 (10,40)	0.449 (11,40)	.003 (0,075)	0.015 (0,38)	.2028 (5,151)	.2026 (5,146)	.010 (0,25)
6	0.442 (11,24)	0.482 (12,24)	.003 (0,075)	0.015 (0,38)	.2560 (6,502)	.2558 (6,497)	.015 (0,38)
8	.544 (13,81)	.588 (14,81)	.003 (0,075)	.020 (0,51)	.3732 (9,479)	.3730 (9,470)	.015 (0,38)
10	.623 (15,82)	.670 (16,82)	.003 (0,075)	.025 (0,64)	.4791 (12,169)	.4789 (12,164)	.015 (0,38)

Table 2: Grip lengths
Dimensions in Inch (millimeters)

Grip Code	Grip Range		Permissible grip over lap	Diameter 5 overall Length Ref
	Max	Min		
02	.125 (3,18)	.077 (1,96)	.133 (3,38)	.451 (11,46)
03	.187 (4,75)	.126 (3,20)	.195 (4,95)	.513 (13,03)
04	.250 (6,35)	.188 (4,78)	.258 (6,55)	.576 (14,63)
05	.312 (7,92)	.251 (6,38)	.320 (8,13)	.638 (16,21)
06	.375 (9,53)	.313 (7,95)	.383 (9,73)	.701 (17,81)
07	.437 (11,10)	.376 (9,55)	.445 (11,30)	.763 (19,38)
08	.500 (12,70)	.438 (11,13)	.508 (12,90)	.826 (20,98)
09	.562 (14,27)	.501 (12,73)	.570 (14,48)	.888 (22,56)
10	.625 (15,88)	.563 (14,30)	.633 (16,08)	.951 (24,16)
11	.687 (17,45)	.626 (15,90)	.695 (17,65)	1.013 (25,73)
12	.750 (19,05)	.688 (17,48)	.758 (19,25)	---
13	.812 (20,62)	.751 (19,08)	.820 (20,83)	---
14	.875 (22,23)	.813 (20,65)	.883 (22,43)	---
15	.937 (23,80)	.876 (22,25)	.945 (24,00)	---
16	1.000 (25,40)	.938 (23,83)	1.008 (25,60)	---
17	1.062 (26,97)	1.001 (25,43)	1.070 (27,18)	---
18	1.125 (28,58)	1.063 (27,00)	1.133 (28,78)	---
19	1.187 (30,15)	1.126 (28,60)	1.195 (30,35)	---
20	1.2500 (31,75)	1.188 (30,18)	1,258 (31,95)	---

(Continued)

Table 2: Grip lengths (Concluded)

Dimensions in Inch (millimeters)

Grip Code	Grip Range		Permissible grip over lap	Diameter 6 Overall Length Ref	Diameter 8 Overall Length Ref	Diameter 10 Overall Length Ref
	Max	Min				
02	.125 (3,18)	.084 (2,13)	.133 (3,38)	.482 (12,24)	---	---
03	.187 (4,75)	.126 (3,20)	.195 (4,95)	.544 (13,82)	.649 (16,48)	---
04	.250 (6,35)	.188 (4,78)	.258 (6,55)	.607 (15,42)	.711 (18,06)	.792 (20,12)
05	.312 (7,92)	.251 (6,38)	.320 (8,13)	.669 (16,99)	.774 (19,66)	.854 (21,69)
06	.375 (9,53)	.313 (7,95)	.383 (9,73)	.732 (18,59)	.836 (21,23)	.917 (23,29)
07	.437 (11,10)	.376 (9,55)	.445 (11,30)	.794 (20,17)	.899 (22,83)	.979 (24,87)
08	.500 (12,70)	.438 (11,13)	.508 (12,90)	.857 (21,77)	.961 (24,41)	1.042 (26,47)
09	.562 (14,27)	.501 (12,73)	.570 (14,48)	.919 (23,34)	1.024 (26,01)	1.104 (28,04)
10	.625 (15,88)	.563 (14,30)	.633 (16,08)	.982 (24,94)	1.086 (27,58)	1.166 (29,62)
11	.687 (17,45)	.626 (15,90)	.695 (17,65)	1.044 (26,52)	1.149 (29,18)	1.229 (31,22)
12	.750 (19,05)	.688 (17,48)	.758 (19,25)	1.107 (28,12)	1.211 (30,76)	1.291 (32,79)
13	.812 (20,62)	.751 (19,08)	.820 (20,83)	1.169 (29,69)	1.274 (32,36)	1.354 (34,39)
14	.875 (22,23)	.813 (20,65)	.883 (22,43)	1.232 (31,29)	1.336 (33,93)	1.416 (35,97)
15	.937 (23,80)	.876 (22,25)	.945 (24,00)	---	1.399 (35,53)	1.479 (37,57)
16	1.000 (25,40)	.938 (23,83)	1.008 (25,60)	---	---	1.541 (39,14)
17	1.062 (26,97)	1.001 (25,43)	1.070 (27,18)	---	---	1.604 (40,74)
18	1.125 (28,58)	1.063 (27,00)	1.133 (28,78)	---	---	1.666 (42,32)
19	1.187 (30,15)	1.126 (28,60)	1.195 (30,35)	---	---	1.729 (43,92)
20	1.2500 (31,75)	1.188 (30,18)	1,258 (31,95)	---	---	1.791 (45,49)

Table 3: Oversize dimensions (1st Oversize) 1/64

Dimensions in Inch (millimeters)

Diameter code	ØA Theo	ØA1	ØD		F	
			Max	Min	Max	Min
5	Use next nominal diameter (diameter 6)					
6X	.3277 (8,324)	.2929 (7,440)	.2026 (5,146)	.2016 (5,121)	.0295 (0,749)	.0263 (0,668)
8X	.4283 (10,879)	0.393 (10,01)	.2651 (6,734)	.2641 (6,708)	.0227 (0,577)	.0195 (0,495)
10X	.5361 (13,617)	0.501 (12,73)	.3276 (8,321)	.3266 (8,296)	.0234 (0,594)	.0198 (0,503)

Table 3: Oversize dimensions (1st Oversize) 1/64(concluded)

Dimensions in Inch (millimeters)

Size code	ØV		H	ØJ Min	K Max	M
	Max	Min				
5	Use next nominal diameter (diameter 6)					
6X	.2560 (6,502)	.2558 (6,497)	.052 (1,32)	.254 (6,45)	0.470 (11,95)	0.510 (12,95)
8X	.3732 (9,479)	.3730 (9,474)	.068 (1,73)	.332 (8,43)	.576 (14,63)	.620 (15,75)
10X	.4791 (12,169)	.4789 (12,164)	.087 (2,21)	.410 (10,41)	.730 (18,54)	.777 (19,74)

Table 4: Oversize Dimensions (2nd Oversize) 1/32

Dimensions in Inch (millimeters)

Diameter code	Preferred Fastener
5	Use diameter 6X
6	Use EN6114K3A-** or ASNA2657-3-**Y
8	Use EN6114K4-()Y
10	Use EN6114K5-()Y

Table 5: Material and surface treatment

Material code	Material		Surface Treatment		Lubricant	
	Sleeve	Core-Bolt	Sleeve	Core-Bolt	Sleeve	Core-Bolt
K	A286 Corrosion Resisting Steel chemical composition per AMS5732 or AMS5737 (1207 MPa) Tensile minimum	6AL-4V Titanium Alloy In accordance with AMS4928 or AMS4967 To 95 KSI (655 MPa) shear strength minimum.	Aluminium Coat in accordance with EN4473 type II	Dry Film Lube per AS5272	Cetyl Alcohol lube per EN6117 (Optional)	Cetyl Alcohol lube per EN6117 (Optional)

Table 6: Mechanical Characteristics

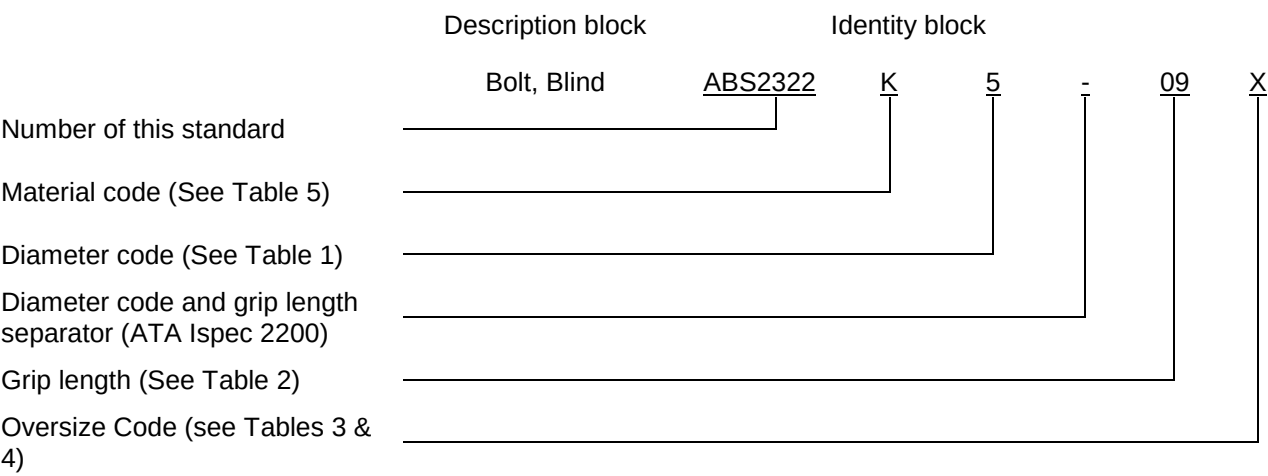
Diameter Code	Min. Single Shear Strength (N)	Min. Tensile Strength (N)	Max. Fatigue Load (N)	Min. Preload (N)
5	8 919	5 250	2290	2224
6	11 966	7 100	3560	3560
8	20 684	13345	5560	6670
10	32 472	22240	8540	11120

Table 7: Mass (Installed) in Grams

Grip Code	5	6	8	10
02	0,86	1,27	---	---
03	0,98	1,43	2,93	---
04	1,10	1,59	3,20	5,67
05	1,21	1,75	3,47	6,10
06	1,33	1,91	3,74	6,53
07	1,45	2,06	4,01	6,95
08	1,57	2,22	4,28	7,38
09	1,68	2,38	4,55	7,80
10	1,80	2,54	4,82	8,23
11	1,92	2,70	5,10	8,65
12	---	2,86	5,37	9,08
13	---	3,02	5,64	9,50
14	---	3,18	5,91	9,93

4 Designation

This type of standard shall be designated according to the philosophy of the following example:



5 Marking

Each part shall be permanently and legibly marked as per EN2424, Style F. and additional oversize code (e.g. X), when necessary. Diameters 8 and 10 are marked with a square or square outile to designate intermediate head style. The marking shall be recessed with maximum depth of 0,25 mm.

6 Technical specification

ABS2320. Type 2, Style B, Class 2, Code B, Subclass 2.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 04/18		New standard
2 09/18	Section 2 Section 3.1 Section 3.2, Figure 1 Tables 1, 3 and 4 Table 5 Table 6	Normative references updated. Ammended to include ISO2768-1 and ISO2768-2 and to link mass properties to table 7. Figure 1 updated to align dimensions J, M , ØV and Z (formerly ØBD, BC, ØG and J respectively) to ABS2320. Dimensions G max and ØK added. Thread locking feature updated. Table 4 amended to align with diameter codes 5 , 6 and 6X. Dimensions G max and K max introduced. Surface treatment and lubricant updated. Min. Double Shear Strength requirement replaced by Min. Single Shear Strength. Min. Tensile Strength and Min. Preload limits updated.
3 05/19	Table 1 Tables 3, 4, 6 and 7	Dimesion F, K and M corrected in accordance with installation test. Diameters 8 and 10 added. Diameters 8 and 10 added.